GEORGE NGUGI

Software Developer | Full-Stack & Mobile Application Development

Nairobi, Kenya | geomaina04@gmail.com
Portfolio: https://georgeportfoliowebsite.vercel.app/ | GitHub: https://github.com/georgengugi-04

PROFESSIONAL SUMMARY

Final-year BSc Information Technology student at KCA University with practical experience building web and mobile applications. Skilled in JavaScript, TypeScript, React, Next.js, Node.js, Flutter and Firebase, with hands-on experience building APIs, integrating databases, implementing authentication and deploying applications. Has delivered solutions in agriculture, cultural technology and business systems, and is particularly interested in reliable, scalable backend and full-stack engineering.

TECHNICAL SKILLS

Programming: JavaScript, TypeScript, Dart, PHP
Frontend: HTML5, CSS3, React.js, Next.js, Tailwind CSS
Backend: Node.js, Express.js, REST APIs
Mobile: Flutter
Databases: PostgreSQL, MySQL, Firebase Firestore, Supabase
Tools & Platforms: Git, GitHub, Docker, Postman, Vercel, Railway, Cloudflare
Practices: Authentication, API integration, database design, automated testing, debugging, responsive UI

PROFESSIONAL EXPERIENCE

JHUB AFRICA — Mobile Developer (Attachment) *June 2026 – September 2026*

- Worked on GreenTrack, an agricultural technology platform supporting farmers with access to agricultural information.
- Contributed to Flutter mobile development and Firebase integration, including authentication and Firestore data management.
- Integrated external services and explored weather and crop-health information within the platform.
- Took part in testing, troubleshooting and improvements based on user feedback.
- Joined field validation in Subukia, Nakuru County, engaging farmers and other stakeholders to gather feedback on the solution.

SELECTED PROJECTS

PAMOJA — Reliable Stock Reservation Service *Backend Engineering Assessment*

Node.js, TypeScript, Express, PostgreSQL, Docker | github.com/georgengugi-04/pamoja-stock-reservation

- Built a stock reservation API using PostgreSQL transactions and an atomic conditional UPDATE that prevents overselling under concurrent requests.
- Implemented idempotency keys backed by a database unique constraint, so retries and duplicate requests never reserve stock twice.
- Implemented reservation release and expiry with row locking, restoring stock exactly once even when release and expiry race.
- Wrote 64 automated tests (Vitest, Supertest) against a real PostgreSQL database, including concurrent-request and race-condition tests.
- Documented the design, trade-offs and limitations in a technical note; database run locally via Docker Compose.

GreenTrack — Agricultural Technology Platform *Mobile & Web*

Flutter, Dart, Firebase

- Developed a garden-to-table tracking platform to support farmers and improve access to agricultural information.
- Implemented crop and garden tracking, harvest records, nutrition calculations and community trading features.
- Integrated weather and crop-health services; validated with farmers, chefs and stakeholders in Subukia, Nakuru County.

EartGalla — Kenyan Art & Culture Platform *Founder & Developer*

Next.js | https://eartgalla.vercel.app/

- Building a digital platform that showcases Kenyan art, culture, heritage and creative communities.
- Developing web experiences with Next.js; exploring digital galleries, artist discovery and marketplace functionality.

BeefTrace — Livestock Traceability Platform *Software Developer*

Web technologies and backend integrations | https://beef-gamma.vercel.app/

- Contributed to a livestock traceability platform covering digital livestock records, animal identification and QR-based traceability.
- Worked on data management and user-friendly interfaces for livestock information.

EDUCATION

KCA UNIVERSITY — BSc Information Technology *Expected graduation: November 2026*

ST PAUL'S AMANA BOYS — KCSE *Mean Grade: C+*

OAKLANDS JUNIOR SCHOOL — KCPE *318 marks*

PROFESSIONAL INTERESTS

Backend engineering • Full-stack development • Mobile development • Database design • APIs and cloud technologies • System reliability

REFERENCES

Available upon request.